

Timothy Keyes

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Education

MD | Stanford University School of Medicine, Stanford, CA | *Expected 2028* - MD/PhD program

PhD in Cancer Biology | Stanford University School of Medicine, Stanford, CA | *Conferred* - Advisors: Kara L. Davis, DO & Garry P. Nolan, PhD - Focus: machine learning applied to single-cell data in pediatric leukemia

MS in Biomedical Data Science | Stanford University School of Medicine, Stanford, CA | *Conferred* - Thesis readers: Robert Tibshirani, PhD & Trevor Hastie, PhD

BA in Psychology and Neuroscience | Princeton University, Princeton, NJ | *Conferred* - Summa cum laude - Concentrations in Neurobiology and Quantitative/Computational Neuroscience - Phi Beta Kappa

Year Abroad | St Edmund Hall - University of Oxford, Oxford, UK | *Completed* - Concentrations: Biochemistry, Experimental Psychology

Experience

Senior Data Scientist | Stanford Health Care - Technology & Digital Solutions, Stanford, CA | *Current* - Clinical AI platform (ChatEHR) - LLM-powered automations for clinical workflows - Continuous monitoring of deployed AI systems

Data Science Mentor (Contract) | Posit, PBC, Remote | *Current* - Mentor for Posit Academy cohorts learning R and Python

Graduate Research Assistant | Davis and Nolan Laboratories, Stanford University School of Medicine, Stanford, CA | *Former* - PhD research in computational cancer biology - Machine learning on single-cell data in pediatric leukemia - Developed several R/Python packages for single-cell data analysis

Data Scientist & Founding Member | Medical Student Pride Alliance (MSPA), Remote | *Former* - Data analysis and visualization for strategic decision-making - Mentored student analysts in Python and R

Laboratory Technician | Kipnis Laboratory, University of Virginia, Charlottesville, VA | *Former* - Contributed to the novel characterization of meningeal lymphatic vessels in the mouse and human CNS - Led 3D computational modeling of meningeal lymphatics

Undergraduate Research Assistant | Gould Laboratory, Princeton University, Princeton, NJ | *Former* - Investigated effects of physical activity and norepinephrine signaling on astrocyte and microglial morphology in the adult rat hippocampus

Publications

- **Timothy Keyes**[†], Jane Wang[†], April S. Liang, Stephen P. Ma, Jason Shen, Jerry Liu, Nerissa Ambers, Abby Pandya, Rita Pandya, Jason Hom, Natasha Steele, Jonathan H. Chen, Kevin Schulman. Deployment and Evaluation of an EHR-integrated, Large Language Model-Powered Tool to Triage Surgical Patients. *arXiv* (2026).
- **Timothy Keyes**[†], Shyon Parsa[†], Dev Dash, Danton Char, Michelle M. Mello, Alison Callahan, Margaret Ann Smith, Sinjin Lee, Thomas Wang, Heidi Salisbury, Shinichi Goto, Vicki Parikh, Kenneth W. Mahaffey, Michael Salerno, Euan A. Ashley, Nigam H. Shah, and Sneha S. Jain. Designing Clinically Useful AI: A Blueprint for Impact. *NEJM AI* (2026).

- **Timothy Keyes**[†], Tim Ellis-Caleo[†], Nerissa Ambers, Faraah Bekheet, Wen-wai Yim, Nikesh Kotecha, Nigam H. Shah, Joel Neal. Development, Evaluation, and Deployment of a Multi-Agent System for Thoracic Tumor Board. *arXiv* (2026).
- Francois Grolleau, **Timothy Keyes**[‡], April S. Liang[‡], Stephen P. Ma, Thomas Lew, Tridu R. Huynh, Natasha Steele, Philip Chung, Paige Qin, Gowri Chandra, Stephanie F. Wang, Evan Mullen, Lauren Carpenter, Mita Hoppenfeld, Matthew Morrin, Baffour A. Kyerematen, Nerissa Ambers, Nikesh Kotecha, Emily Alsentzer, Jason Hom, Nigam H. Shah, Kevin Schulman, Jonathan H. Chen. MedAgentBrief for Hospital Course Summarization: Safety, Use, and Discharge Documentation Burden. *medRxiv* (2026).
- Nigam H. Shah, Nerissa Ambers, Abby Pandya, **Timothy Keyes**, Juan M. Banda, Srikar Nallan, Carlene Lugtu, Artem A. Trotsyuk, Suhana Bedi, Alyssa Unell, Miguel Fuentes, Francois Grolleau, Sneha S. Jain, Jonathan Chen, Dev Dash, Danton Char, et al.. Adoption and Use of LLMs at an Academic Medical Center. *arXiv* (2026).
- Yuxuan Liu, Haowen Jiang, Jingjing Liu, Lucille Stuan, Milton J Merchant, Astraea Jager, Abhishek Koladiya, Ti-Cheng Chang, Pablo Domizi, Jolanda Sarno, Ao Wang, **Timothy Keyes**, Dorra Jedoui, Jodie Meng, Felix Hartmann, Ruida Hou, Carol Fries, Chiara Pirillo, Qingsong Gao, Ilaria Iacobucci, Sean C Bendall, Min Huang, Norman J Lacayo, Kathleen M Sakamoto, Charles G Mullighan, Mignon L Loh, Jiyang Yu, Jun J Yang, Jiangbin Ye, and Kara L Davis. DHODH as a Targetable Metabolic Achilles' Heel for chemo-resistant B-ALL. *Blood* (2026).
- **Timothy Keyes**[†], Alison Callahan[†], Abby S. Pandya[†], Nerissa Ambers, Juan M. Banda, Miguel Fuentes, Carlene Lugtu, Pranav Masariya, Srikar Nallan, Connor O'Brien, Thomas Wang, Emily Alsentzer, Jonathan H. Chen, Dev Dash, Matthew A. Eisenberg, Patricia Garcia, Nikesh Kotecha, Anurang Revri, Michael A. Pfeffer, Nigam H. Shah, Sneha S. Jain. Monitoring Deployed AI Systems in Health Care. *arXiv* (2025).
- **Timothy Keyes**[†], Erin Craig[†], Jolanda Sarno, Jeremy P. D'Silva, Pablo Domizi, Maxim Zaslavsky, Albert Tsai, et al.. Annotation-free discovery of disease-relevant cells in single-cell datasets. *Science Advances* (2025).
- Eugenia Miranti, **Timothy Keyes**, Alvaro Ayala, Nerissa Ambers, Gina Newman, Elmer de Leon, Erika Paola Viana-Cardenas, Wajeeha Tariq, Mindy Sampson, and Jorge L. Salinas. Use of a large language model integrated within the electronic medical record for the evaluation of surgical site infections. *Infection Control & Hospital Epidemiology* (2025).
- Francois Grolleau, Emily Alsentzer, **Timothy Keyes**, Philip Chung, Akshay Swaminathan, Asad Aali, Jason Hom, Tridu Huynh, Thomas Lew, April S. Liang, Weihan Chu, Natasha Z. Steele, Christina F. Lin, Jingkun Yang, Kameron C. Black, Stephen P. Ma, Fateme N. Haredasht, Nigam H. Shah, Kevin Schulman, Jonathan H. Chen. MedFactEval and MedAgentBrief: A Framework and Workflow for Generating and Evaluating Factual Clinical Summaries. *Biocomputing 2026: Proceedings of the Pacific Symposium* (2025).
- Guillermo Rodriguez-Nava, **Timothy Keyes**, Nerissa Ambers, Eugenia Miranti, Erika Paola Viana-Cardenas, Wajeeha Tariq, Mindy Marie Sampson, Jorge Luis Salinas. Using secure artificial intelligence agents integrated within the electronic medical record for the evaluation of blood culture appropriateness. *Infection Control & Hospital Epidemiology* (2025).
- Shyon Parsa, **Timothy Keyes**, Dev Dash, Michelle Mello, Heidi Salisbury, Alison Callahan, Shinichi Goto, Michael Salerno, Victoria Parikh, Kenneth Mahaffey, Euan Ashley, Nigam Shah, Sneha Jain. Target Product Profile to Evaluate the Clinical Utility, Financial Impact, and Ethical Implications of an AI-Based HCM Detection Model. *Circulation* (2025).
- Suhana Bedi, Hejie Cui, Miguel Fuentes, Alyssa Unell, Michael Wornow, Juan M. Banda, Nikesh Kotecha, **Timothy Keyes**, Yifan Mai, Mert Oez, Hao Qiu, Shrey Jain, Leonardo Schettini, Mehr Kashyap, Jason Alan Fries, et al.. Holistic evaluation of large language models for medical tasks with MedHELM. *Nature Medicine* (2025).
- Tomasz Wlodarczyk, Aaron Lun, Diana Wu, Minyi Shi, Xiaofen Ye, Shreya Menon, Shushan Toneyan, Kerstin Seidel, **Timothy Keyes**, et al.. Epiregulon: Single-cell transcription factor activity inference to predict drug response and drivers of cell states. *Nature Communications* (2025).
- **Timothy Keyes**[†], William J. Hutchison[†], Helena L. Crowell, Jacques Serizay, Charlotte Soneson, Eric S. Davis, Noriaki Sato, Lambda Moses, Boyd Tarlinton, Abdullah A. Nahid, Miha Kosmac, Quentin Clayssen, Victor Yuan, Wancen Mu, Ji-Eun Park, Izabela Mamede, Min Hyung Ryu, Pierre-Paul Axisa, Paulina Paiz, Chi-Lam Poon,

- Ming Tang, Raphael Gottardo, Martin Morgan, Stuart Lee, Michael Lawrence, Stephanie C. Hicks, Garry P. Nolan, Kara L. Davis, Anthony T. Papenfuss, Michael I. Love, Stefano Mangiola. The tidyomics ecosystem: enhancing omic data analyses. *Nature Methods* (2024).
- Alea Delmastro, Candace Liu, Xiao-Wen Ding, Serena Tan, Inna Averbukh, Marc Bosse, **Timothy Keyes**, et al.. IFN-gamma-Expressing Myeloid Cells Localize within Lipoproteinosis during Drug-Associated Pulmonary Alveolar Proteinosis occurring in Systemic Juvenile Idiopathic Arthritis. *bioRxiv* (2024).
 - Darnell K. Adrian Williams, Briana Christophers, **Timothy Keyes**, Rachit Kumar, Michael C. Granovetter, Alexandria Adigun, Justin Olivera, Jehron Pura-Bryant, Chynna Smith, Chiemeka Okafor, Mahlet Shibire, Dania Daye, Myles H. Akabas. Sociodemographic factors and research experience impact MD-PhD program acceptance. *JCI Insight* (2024).
 - **Timothy Keyes**, Abhishek Koladiya, Yu-Chen Lo, Garry P. Nolan, Kara L. Davis. tidytof: a user-friendly framework for scalable and reproducible high-dimensional cytometry data analysis. *Bioinformatics Advances* (2023).
 - Michael Albert Gisoni, **Timothy Keyes**, Shana Zucker, Deila Bumgardner. Teaching LGBTQ+ health, a web-based faculty development course: program evaluation study using the RE-AIM framework. *JMIR Medical Education* (2023).
 - Yu-Chen Lo, Yuxuan Liu, Marte Kammersgaard, Abhishek Koladiya, **Timothy Keyes**, Kara L. Davis. Single-cell technologies uncover intra-tumor heterogeneity in childhood cancers. *Seminars in Immunopathology* (2023).
 - **Timothy Keyes**, Astraea Jager, Mason Krueger, Sylvia Plevritis, Robert Tibshirani, Richard Aplenc, et al.. Improved Relapse Prediction in Pediatric Acute Myeloid Leukemia By Deconvolving Lineage-Specific and Cancer-Specific Features in Single-Cell Data. *Blood* (2022).
 - **Timothy Keyes**, Shana Zucker, Teddy G. Goetz, Justin L. Jia, Samuel R. Bunting, Mitchell R. Lunn, Leslee L. Subak. Sexual and gender minority identity disclosure from undergraduate to graduate medical education: perceptions of professional Outness among Medical Students. *Annals of LGBTQ Public and Population Health* (2022).
 - Yu-Chen Lo, **Timothy Keyes**, Astraea Jager, Jolanda Sarno, Pablo Domizi, Ravindra Majeti, Kathleen M. Sakamoto, Norman Lacayo, Charles G. Mullighan, Jeffrey Waters, Bitu Sahaf, Sean C. Bendall, Kara L. Davis. CytoIn enables integrated analysis of public mass cytometry datasets using generalized anchors. *Nature Communications* (2022).
 - **Timothy Keyes**, Pablo Domizi, Yu-Chen Lo, Garry P. Nolan, Kara L. Davis. A cancer biologist's primer on machine learning applications in high-dimensional cytometry. *Cytometry Part A* (2020).
 - Connie H. Lin, Hye Sun Kuehn, Timothy J. Thauland, Christine M. Lee, Suk See De Ravin, Harry L. Malech, **Timothy Keyes**, Astraea Jager, Kara L. Davis, Maria I. Garcia-Lloret, Sergio D. Rosenzweig, Manish J. Butte. Progressive B Cell Loss in Revertant X-SCID. *Journal of Clinical Immunology* (2020).
 - Samuel R. Bunting, Sarah S. Garber, Robert H. Goldstein, Timothy D. Ritchie, Tamzin J. Batteson, **Timothy Keyes**. Student Education About Pre-Exposure Prophylaxis (PrEP) Varies Between Regions of the United States. *Journal of General Internal Medicine* (2020).
 - Teddy G. Goetz, Shana Zucker, **Timothy Keyes**, Michael Gisoni. Medical Student Pride Alliance: The First National LGBTQ+ Medical Student Affinity Organization. *Medical Education* (2020).
 - Justin L. Jia, Marija Kamceva, **Timothy Keyes**. Navigating Controversy: A Critical Element of Medical Education. *Academic Medicine* (2018).
 - Antoine Louveau, Igor Smirnov, **Timothy Keyes**, Jacob D. Eccles, Sherin J. Rouhani, J. David Peske, Noel C. Derecki, David Castle, James W. Mandell, Kevin S. Lee, Tajie H. Harris, Jonathan Kipnis. Structural and functional features of central nervous system lymphatic vessels. *Nature* (2015).

Awards & Honors

- **Integrated Strategic Plan Star Award (ChatEHR Team)**, Stanford Medicine (2025)
- **Health Catalyst Ignite Award**, Health Catalyst (with Stanford Hospital Quality) (2025)
- **Point Graduate Student Scholarship**, Point Foundation (2020)

- **Ruth L. Kirschstein Pre-doctoral NRSA (F31)**, National Institutes of Health (2019)
- **Community Impact Award**, Stanford University (2019)
- **Integrated Strategic Plan Star Award**, Stanford Medicine (2019)
- **LGBT Workforce Junior Leadership Award**, Building the Next Generation of Academic Physicians (BNGAP) (2019)
- **Award for Excellence in Promotion of Diversity and Societal Citizenship**, Stanford University School of Medicine (2018)
- **Award for Academic Excellence**, Princeton Neuroscience Institute (2014)
- **Phi Beta Kappa**, Princeton University (2013)
- **Howard Crosby Warren Junior and Senior Prizes in Psychology**, Princeton University (2013 & 2014)
- **Shapiro Prize for Academic Excellence**, Princeton University (2011 & 2012)

Media & Press

- Meet 4 developers leading the way with AI agents. Featured Article — Microsoft Source (2025)
- Microsoft Opens AI ‘Store’ for Healthcare Developers. Featured Article — The New Stack (2025)
- Stanford Medicine and the healthcare agent orchestrator: Satya Nadella at Microsoft Build 2025. Keynote Feature — Microsoft Build 2025 (2025)
- U.S. medical schools boost LGBTQ students, doctor training. Featured Interview — Associated Press, New York Times, Washington Post (2020)
- Meet the Point Foundation’s Class of 2020. Featured Bio — The Advocate (2020)
- Why being a programmer will make me a better doctor. Blog Post — Stanford Medicine Scope Blog (2020)
- Is becoming a physician-scientist worth sacrificing work-life balance?. Blog Post — Stanford Medicine Scope Blog (2020)
- Advice for LGBTQ+ medical school applicants. Blog Post — Stanford Medicine Scope Blog (2019)
- Finding a lab that feels like home. Blog Post — Stanford Medicine Scope Blog (2019)

Skills

Clinical AI & LLM Systems: EHR-integrated AI applications, agentic and multi-agent workflow design, LLM evaluation, retrieval-augmented generation (RAG), clinical summarization, post-deployment monitoring

Machine Learning & Statistics: supervised learning, calibration, bootstrap inference, survival analysis, class imbalance adjustment, clustering, gradient boosting, deep learning, tabular foundation models (tabPFN)

Healthcare Data & Governance: Epic EHR data, FHIR, clinical workflow integration, AI governance, model lifecycle management, clinical operations analytics, human-in-the-loop workflow design

Programming & Data Science: Python, R, SQL, Bash, pandas, scikit-learn, XGBoost, PyTorch, tidyverse, tidy-models, Bioconductor, Shiny

Platforms & Tools: Databricks, Azure OpenAI, LiteLLM, FastAPI, Pydantic, Git, Docker, Epic, MCP, Quarto

Biomedical Domains: clinical machine learning, LLMs in health care, single-cell analysis, mass cytometry, computational cancer biology, pediatric leukemia